

⇒ Deep learning da turli x'l xolatlari uchun NN ning turli narxlar

MLP (Fully Connected NN) ⇒ Tabalar data

CNN ⇒ Images

RNN ⇒ Sequential data

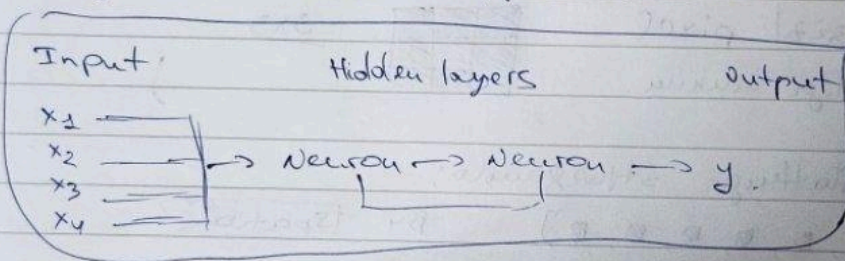
LSTM/GRU ⇒ Long sequences

Transformer ⇒ NLP, Vision

Autoencoder ⇒ Compression

GAN ⇒ Image generation

MLP (Multilayer Perceptron) ⇒ fully connected layerlardan tashkil topgan (classical NN)



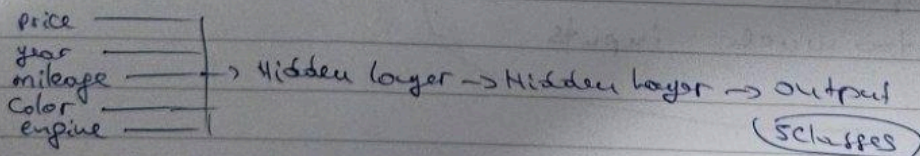
1. Neuron avval weighted sum hisoblaydi.

$$z = w_1x_1 + w_2x_2 + \dots + w_nx_n + b$$

activation function ⇒ $a = f(z)$ $\text{ReLU}(z) = \max(0, z)$

Input ⇒ weighted sum ⇒ Activation ⇒ Output
 $x \Rightarrow w \cdot x + b \Rightarrow \text{ReLU} \Rightarrow a$

Car brand classification



Input: 5 features:

5 ⇒ 256 neurons ⇒ 128 neurons ⇒ 5 neurons

Bir layerdagi har bir neuron keyingi layerdagi barcha neuronlar bilan bog'langan, shuning uchun Fully connected deyiladi.

MLP mino uchun image uchun yaxshi emas?

Masalan: 28×28 grayscale image.

MLP ga berish uchun flatten bōlikni k-k. 1D
[0.2, 0.8, 0.9, 0.5, ..., 0.1] [x₁, x₂, x₃, ..., x_n]

Hatij muammo: pixellar orasidagi relationshipda. Image

Bu yerdə g'izil pixel muvazozdaligi muhim.



3x3

lekin MLP flattenga o'tkazganda:

[0.2, 0.8, 0.9, 0.5, ..., 0.1]

By **Spatial**

relationship ni bilmaydiki

→ Parametrlar juda kam ko'payib ketadi.

$224 \times 224 \times 3$ (RGB) 150528 inputs

1 ⇒ 1D (flatten)

2 ⇒ Spatial relationship

3 ⇒ too much inputs

CNN: Convolutional Neural network
CNN imagedagi local (pattern)ni topadi

CNN ikki qismdan iborat:

Feature extraction

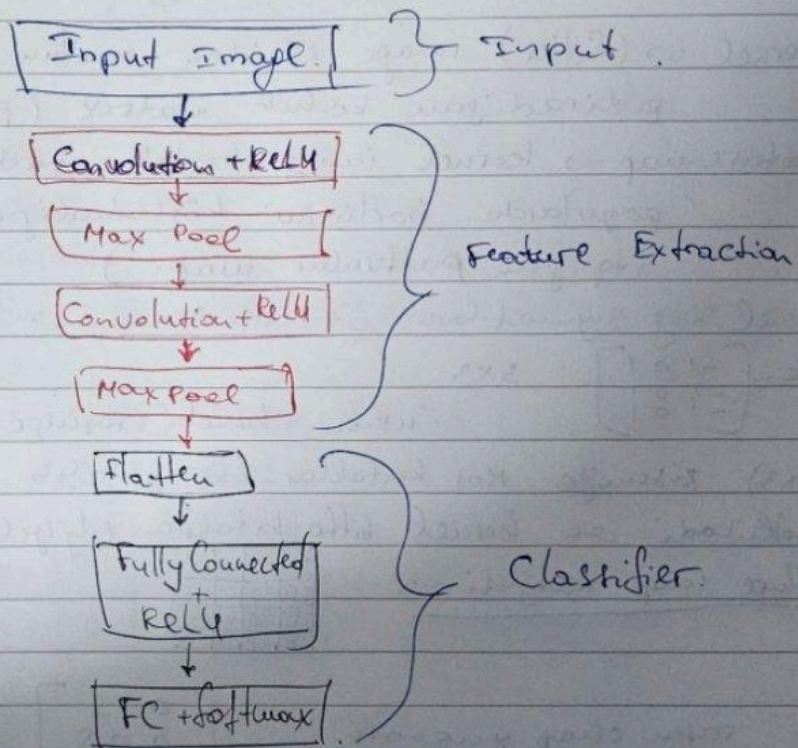


Rasmdan belgilarni topadi.

Classifier



Belgilarga qarab sinf
taulaydi.



INPUT

36 pixel

Image: k_2

$6 \times 6 \times 1$

0	1	8	8	0	0
0	0	8	8	1	0
1	0	8	8	0	0
0	0	8	8	0	1
0	1	8	8	0	0
0	0	8	8	1	0

Convolution $\Rightarrow$

kernel $\Rightarrow$ (filter) image dan ma'lum patterni
gidiradigan kichik matrix (pattern gidiruvchi)

Feature map $\Rightarrow$ kernel image boylab yurib pattern
joylarda borligini ko'rsatadigan natija matritsa
(topilgan patternlar map)

kernel ning qiymatlarini o'ti tanlanishi, slope dan

$$K = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix} \quad 3 \times 3$$

uchin kernel rasme bog'lanadi.

(3x3) zismiga mos kataklar ko'paytirilib hammasi
qo'shiladi va kernel bitta katakka qo'yiladi

Feature map natijasi $\Rightarrow$

25	25	25	25
25	25	25	25
25	25	25	25
25	25	25	25

$$b = 0$$

1- rasmi chap yuqoriti:

$$\begin{bmatrix} 0 & 1 & 8 \\ 0 & 0 & 8 \\ 1 & 0 & 8 \end{bmatrix}$$

mos kataklar
ko'paytirilgan

$$0 \cdot (-1) + 1 \cdot 0 + 8 \cdot 1 + (-1) \cdot 0 + 0 \cdot 0 + 1 \cdot 8 + (-1) \cdot 1 + 0 \cdot 0 + 1 \cdot 8 = 25$$

feature mapning birinchi katapi $\Rightarrow 25$

Feature map 4×4

ReLU:

$$\text{ReLU}(x) = \max(0, x)$$

feature map:

25	25	-25	-26
25	27	-25	-26
25	26	-26	-26
27	25	-26	-25

After ReLU:

25	25	0	0
25	27	0	0
25	26	0	0
27	25	0	0

Pooling (Max Pooling)

Parameters: 2x2 blok
stride=2

blok has safor 2 ta kengligicha soljiydi natija ust tutilmaydi.

1 blok $\Rightarrow (25, 25, 25, 27) \Rightarrow \max \rightarrow 27$
natija 2x2.

27	0
27	0

Flatten:

Poolingdan chiqqan son flatteninge
strokiladi $(27, 0, 27, 0)$

FC layer $\Rightarrow$ NN MLP ning 52.

Softmax: classlarga g'iratis beradi